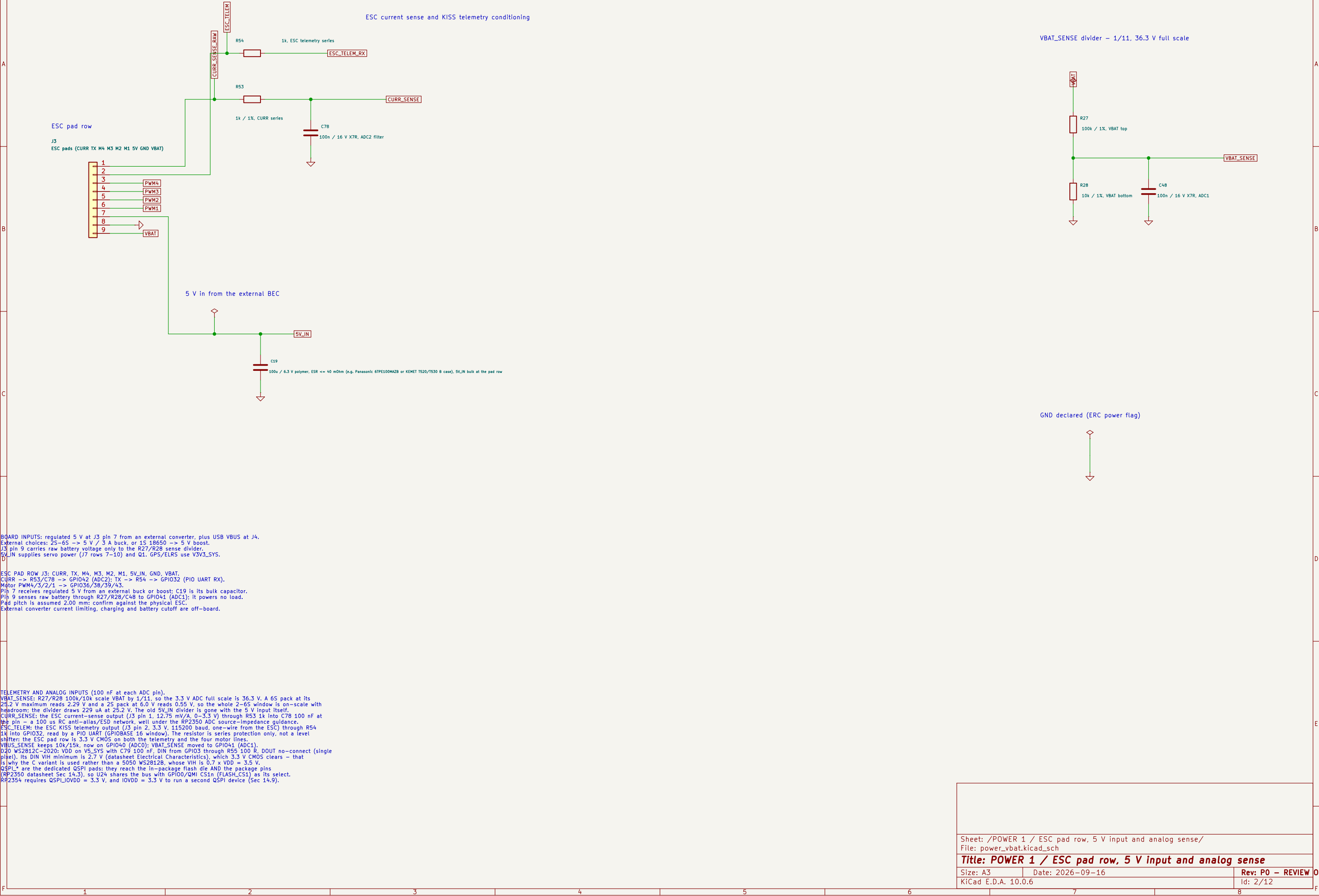


POWER 1 – ESC PAD ROW J3 (NINE PADS), 5 V INPUT FROM AN EXTERNAL BEC, VBAT SENSE, ANALOG SENSE.
Regulated 5 V arrives on J3 pin 7 as 5V_IN; pin 9 brings raw pack VBAT in as a SENSE wire only.



Sheet: /POWER 1 / ESC pad row, 5 V input and analog sense/ File: power_vbat.kicad_sch		
Title: POWER 1 / ESC pad row, 5 V input and analog sense		
Size: A3	Date: 2026-09-16	Rev: P0 – REVIEW ONLY
KiCad E.D.A. 10.0.6		Id: 2/12

POWER 2 – Q1 A03401A P–FET + D1 1N5819WS SCHOTTKY. 5V_IN (external BEC, J3 pin 7) OR USB_VBUS –> V5_SYS.
USB present turns Q1 OFF, so a host can never back-feed the BEC, the servo rows or the ESC.

A

B

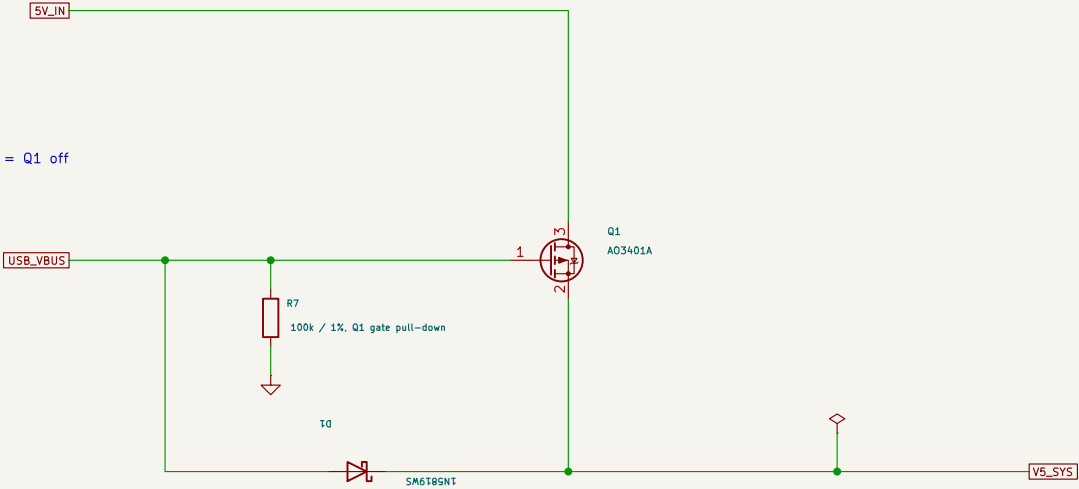
C

D

E

F

BEC path – Q1 P–channel FET, drain 5V_IN, source V5_SYS



USB path – D1 Schottky, the only way USB reaches V5_SYS

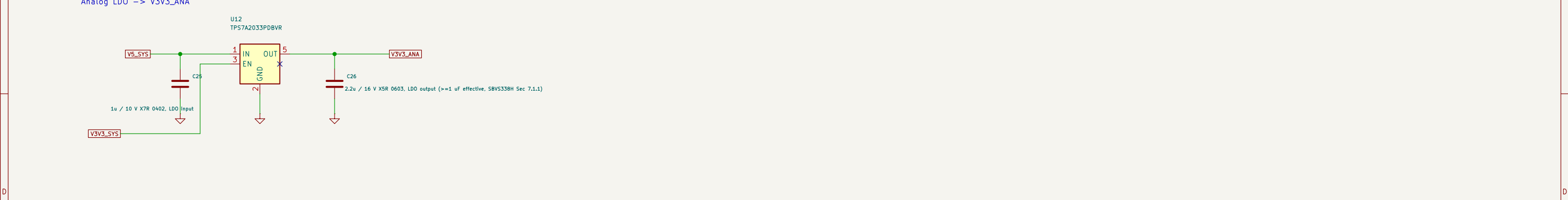
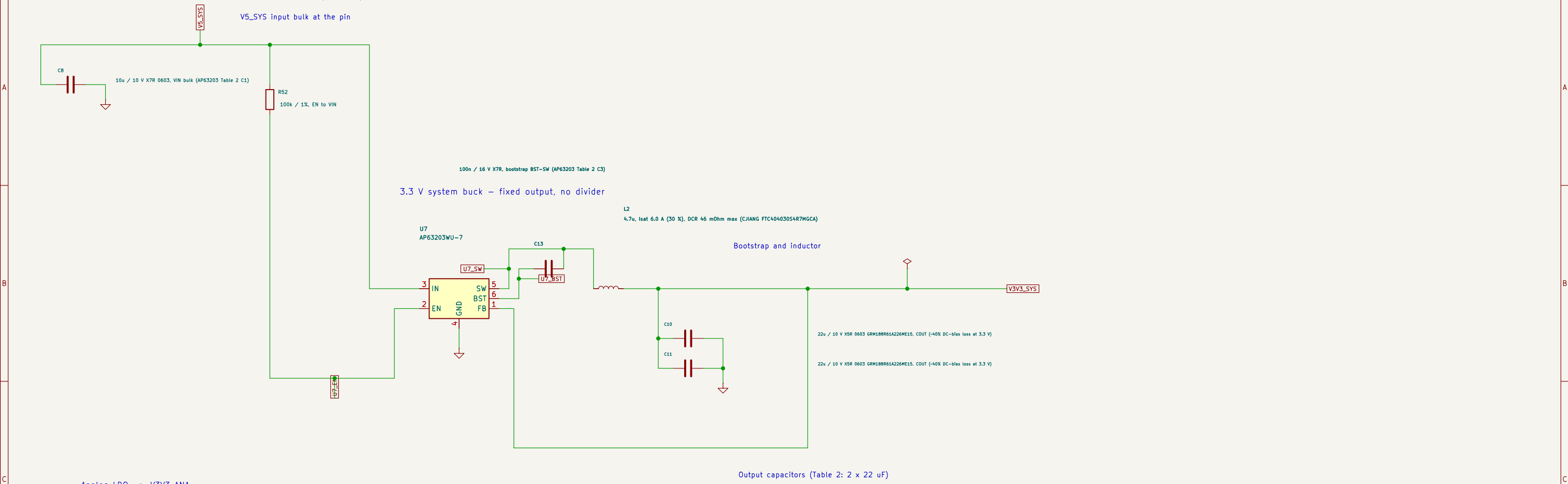
V5_SYS – the two 3.3 V regulators and the status LED, nothing else

5 V INPUT OR: Q1 A03401A and D1 1N5819WS.
Q1 drain = 5V_IN, source = V5_SYS, gate = USB_VBUS. R7 pulls the gate down.
D1 anode = USB_VBUS; cathode = V5_SYS.
Without USB, Q1's channel carries the external supply after its gate discharges.
With USB present, Q1 can be off; D1 and Q1's body diode can share the load.
There is no guaranteed BEC priority or lossless switchover.
C19 (100 uF) is on 5V_IN; C8/C25/C79 are on V5_SYS.
V5_SYS feeds U7 (system buck), U12 (sensor LDO) and the RGB status LED.
GPS/ELRS and other 3.3 V I/O use U7's output; servos stay on external 5V_IN.
Input-only simulation: simulations/power_input.cir. See DESIGN_SPEC.md.

USB_VBUS feeds the D1 Schottky and the Q1 gate on POWER 2 directly; no inrush or current limiting on this sheet, and no data ESD array any more – U8 USBLC6–2SC6 was JLC Extended and is removed (DESIGN_SPEC "Decisions"), so D+/D– go straight from J4 into the 27 R series pair R22/R23, as on the madflight FC3v2. The RP2354B USB pins carry their own on–die ESD structures; a board–level array is what this design gives up, deliberately.

Sheet: /POWER 2 / 5 V OR – BEC P–FET and USB Schottky/ File: power_or.kicad_sch		
Title: POWER 2 / 5 V OR – BEC P–FET and USB Schottky		
Size: A3	Date: 2026–09–16	Rev: P0 – REVIEW ONLY
KiCad E.D.A. 10.0.6	Id: 3/12	

POWER 3 – U7 AP63203WU–7 FIXED 3.3 V BUCK (V5_SYS –> V3V3_SYS) AND U12 TPS7A20 LDO (–> V3V3_ANA)
AVIONICS ONLY: 300 mA continuous / 500 mA start peak, provisional, NO SERVO POWER.



U7 AP63203WU–7, V5_SYS –> V3V3_SYS BUCK (Diodes D541326 Rev. 3–2), 3.8–32 V in, FIXED 3.3 V out, 2 A, 1.1 MHz with +/-6% frequency spread spectrum, 4 ms internal soft–start, 22 uA quiescent, TSOT26. It replaces the TP562913 two–stage stage (U7 + L2 + FB1 + a 0.1 % divider + five 22 uF) at ~8 parts instead of ~25. PASSIVES ARE TABLE 2 “Recommended Component Selections for AP63203” (3.3 V row) and Figure 21, the AP63203/AP63205 typical application, with one documented substitution:

- L 3.9 uH in Table 2 –> 4.7 uH (L2). Sec 10 “Inductor” licenses it explicitly: “It is recommended to select an inductor of approximately 2.2uH to 10uH with a DC current rating of at least 35% higher than the maximum load current... Use a larger inductance for improved efficiency under light load conditions”, and this rail is a light load (~180 mA typical / ~300 mA peak), 4.7 uH is a part the board already sources.
- C1 10 uF –> C8, 10 uF / 25 V X5R 0603, at VIN, and it is the ONLY input capacitor: the C9 100 nF HF bypass that used to sit beside it was an addition beyond Table 2 and is removed.
- C2 2 x 22 uF –> C10 / C11, 22 uF / 10 V X5R 0603 (GRM188R61A226ME15) at the output, exactly the table value.
- C3 100 nF –> C13, bootstrap BST–SW.

FB (pin 1) is a SENSE input on the fixed–output parts and goes straight to V3V3_SYS – Sec 9 “Setting the Output Voltage”: only the adjustable AP63200/AP63201 take a divider (Eq.6). So there is no feedback divider, no 0.1 % resistor pair and no setpoint arithmetic on this board any more: VOUT is the part number.

EN (pin 2): “The EN pin is a high voltage pin and can be directly connected to VIN”, abs max 35 V, threshold 1.18 V rising / 1.10 V falling, internal 1.5 uA pull–up, R52 100k from V5_SYS is that connection with a series element, so a UVLO divider (Eq.1/Eq.2) or a start–delay cap (Eq.3) can be added later without touching copper. OPERATING POINT at VIN = 5.0 V, VOUT = 3.3 V, fsw = 1.1 MHz, Eq.7: dIL = 3.3 x 1.7 / (5.0 x 4.7u x 1.1M) = 0.22 A pk–pk (0.27 A at the ~20% inductance limit), so Eq.8 gives a 0.41 A inductor peak at the 0.3 A peak rail load. On USB, VIN falls to ~4.5 V behind D1 and dIL drops to 0.17 A; the part stays in CCM–adjacent light–load operation either way and neither case comes near the 2 A rating or the inductor.

DERATING CAVEAT: C8 and C10/C11 are the datasheet NOMINAL values and ceramic DC–bias derating is not in them – a 22 uF / 10 V 0603 at 3.3 V keeps roughly 60 %, so the 44 uF of COUT is ~26 uF effective. That is still above the 2 x 22 uF the table asks for in nominal terms and is the same derating the removed stage carried.

L2 CJIANG FTC40403054R7MGCA (LCSC C39676083), the inductor the removed AP63205 stage used as L3 ~ 4.7 uH +/-20%, DCR 41 mOhm typ / 46 max, Irms 4.3 A typ / 4.1 A worst case, Isat 7.0 A typ / 6.0 A worst case, 4.1 x 4.1 x 3.0 mm molded metal–composite, shielded (S2 CJIANG FTC series datasheet Rev 7.0 2025/11/05, Notes 3/4/7; the usable rating is the LESSER of Irms and Isat, so 4.0 A). At a 0.41 A peak that is a 10x margin – enormously over–rated for this rail, and deliberately so: it is a part already on the BOM and on the reel, and it costs 46 mOhm x 0.3 A^2 = 4 mW.

IF THE 4 x 4 x 3.0 mm FOOTPRINT EVER HAS TO SHRINK, the FTC3030200 family (3 x 3 x 2.0, LCSC C7423318 at 2.2 uH) is the same library footprint family and 2.2 uH is still inside the datasheet’s 2.2–10 uH window – at 0.47 A pk–pk ripple.

V3V3_ANA feeds ICM–45686, BMP581, ADXL375 and the RP2354B ADC_AVDD pin (100 nF at each pin on the MCU sheet): VREG_AVDD is filtered from V3V3_SYS instead, so the analog rail carries no core–regulator current. U12 EN (pin 3) is tied to V3V3_SYS itself.

the AP63203 has no power–good output, so the sequencing is done by the rail rather than by a status pin – the LDO enables when the buck output is up and drops with it, which is the same buck–before–LDO ordering PWR_GOOD used to give, at no parts.

What is lost is the “in regulation” qualifier: EN now releases at U12’s own ~0.9 V enable threshold on a rising V3V3_SYS, not at the point a comparator declares the buck settled. For a 10–20 mA sensor rail behind a 4 ms soft–start that is a few hundred microseconds of difference, and the sensors are held in reset by firmware for far longer than that.

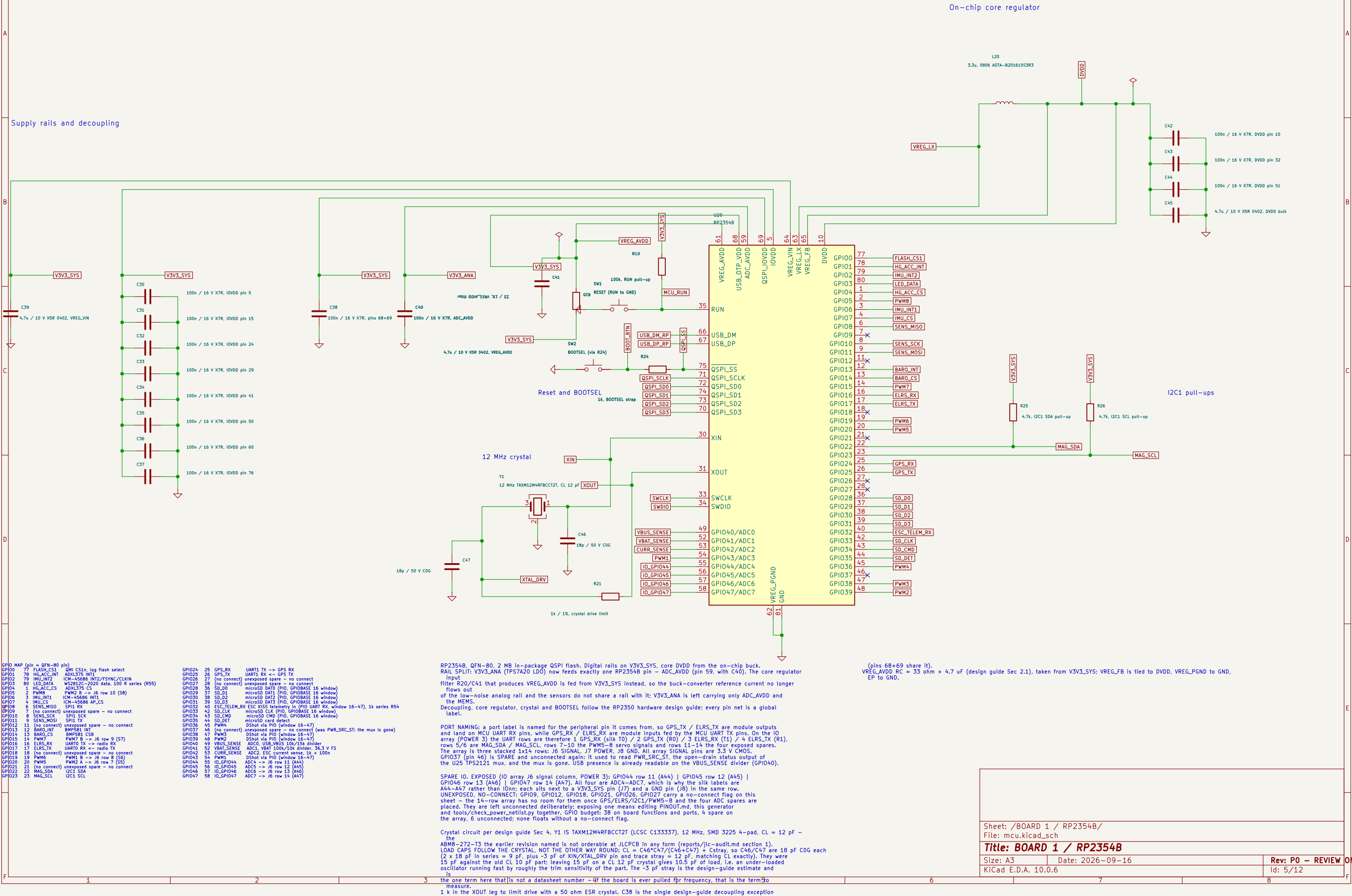
C25 / C26, THE LDO’S OWN CAPACITORS (TPS7A20 datasheet SBV5338h): the part is “stable with 1 uF minimum ceramic output capacitor”, and Sec 7.1.1 says the recommended values already assume “the effective capacitance to decrease by as much as 50 %” – i.e. the 1 uF is an EFFECTIVE floor, not a nameplate one. C26 is therefore 2.2 uF / 16 V X5R in an 0603 (LCSC C23630, JLC Basic), not the 1 uF 0402 it was: an 0402 1 uF biased at 3.3 V has no headroom left against that floor, and the 16 V 0603 keeps well over 1 uF at 3.3 V. C25 stays 1 uF 0402 at the input, where the datasheet asks only for “at least equal to the nominal value” and nothing depends on it.

PASSIVE PACKAGES (DESIGN_SPEC “Physical design”): resistors are 0402, ceramics <= 4.7 uF are 0402 and the 10–22 uF ceramics are 0603. RESISTORS ARE 0402, NOT 0201: JLCPCB has no Basic 0201 resistor of any value, so every 0201 line carried a per–unique–part Extended setup fee – see reports/jlc–audit.md. 0402 makes most of the resistor values Basic and is hand–reworkable, which 0201 is not. THE 0.1 % PAIR IS GONE with the TP562913: the AP63203 is a fixed–output part and has no feedback divider, so no 0.1 % thin–film line is ordered at all. C10/C11 (AP63203 COUT) are 22 uF / 10 V X5R 0603 (GRM188R61A226ME15): at 3.3 V DC bias an X5R 22 uF 0603 keeps roughly 60 % of its nominal value, so the 44 uF nominal is ~26 uF effective. Do not read the effective value off the BOM. The 22 uF 0603 ceramics remain machine–place parts – JLC assembles them, a bench iron will not rework one.

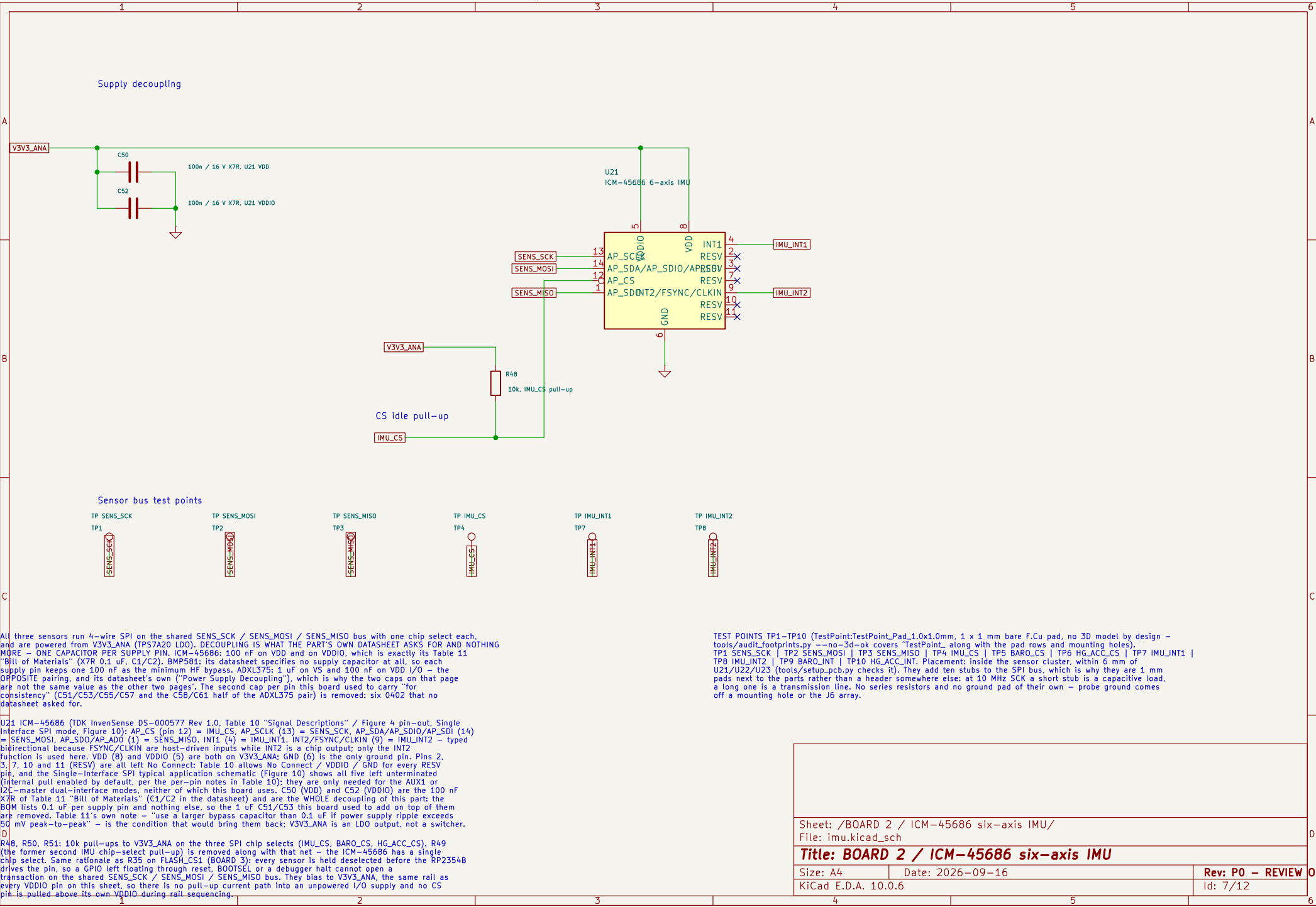
C19, THE 5V_IN BULK, IS 100 uF / 6.3 V POLYMER IN AN EIA–3528–21 (B) CASE. It moved from V5_SYS to 5V_IN with the BEC decision: hold–up belongs on the rail the servo and module rows are on, which is now ahead of the OR–ing FET, and V5_SYS keeps only C8 at the AP63203 input. ESR must stay <= 40 mOhm: Panasonic 6TPE100MAZB or a KEMET T520/T530 B case, 6.3 V on a 5.0 V rail is a 1.26x derating, the normal polymer figure; a tantalum electrolytic would need 2x and would not qualify.

THE ngspice DECKS IN simulations/ STILL MODEL THE OLD TOPOLOGY (a TPS2121 mux fed by an on–board pack buck, and a TP562913–class 3.3 V stage). They were not re–derived for the BEC + P–FET + AP63203 tree, so every number they produce is now history, not evidence – see DESIGN_SPEC “Open items”.

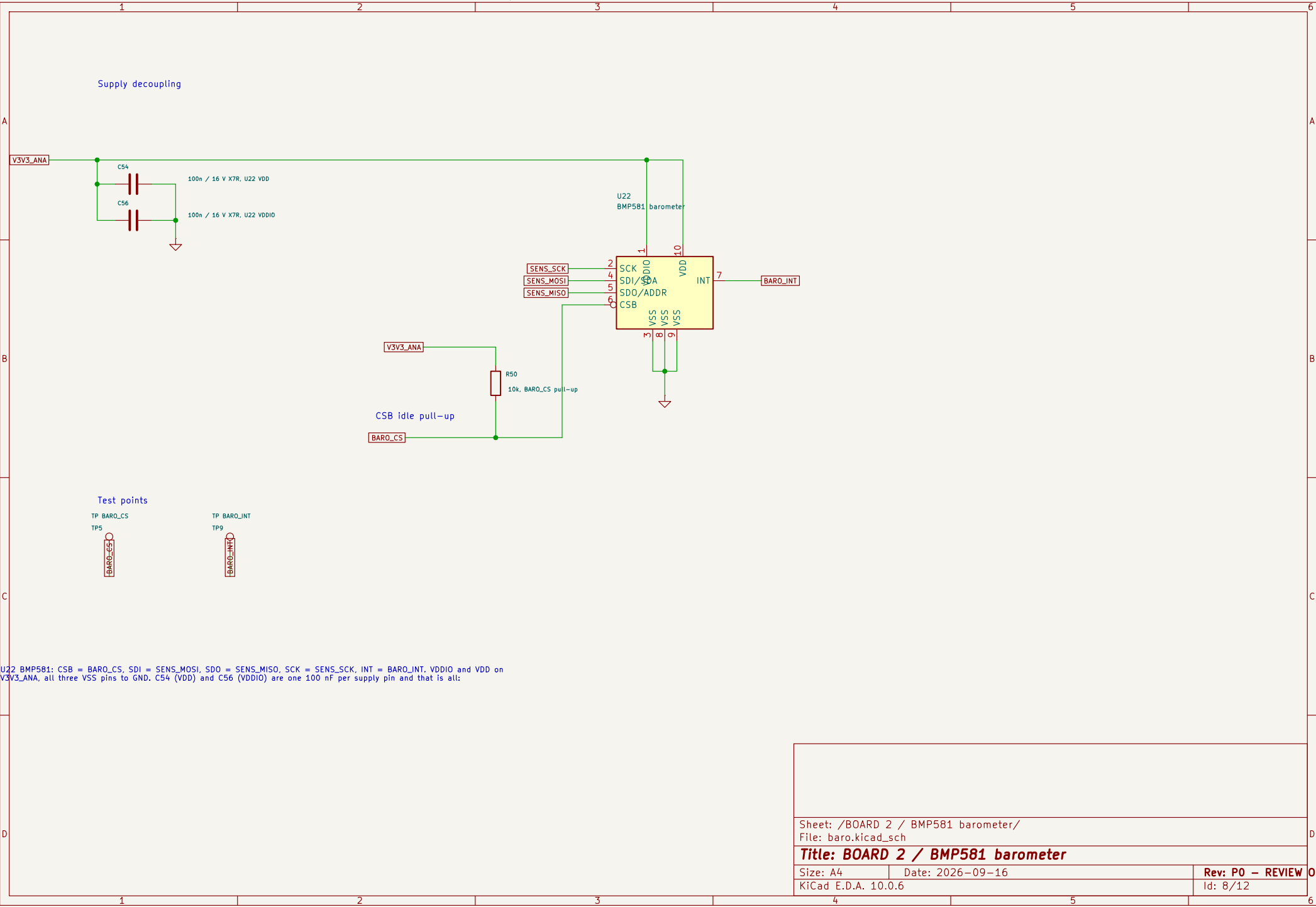
BOARD 1 – RP2354B, QFN–80, 2 MB IN–PACKAGE QSPI FLASH. Digital rails on V3V3_SYS, core DVDD from the on–chip buck. Supply pins fan out on the left into one decoupling group per rail.



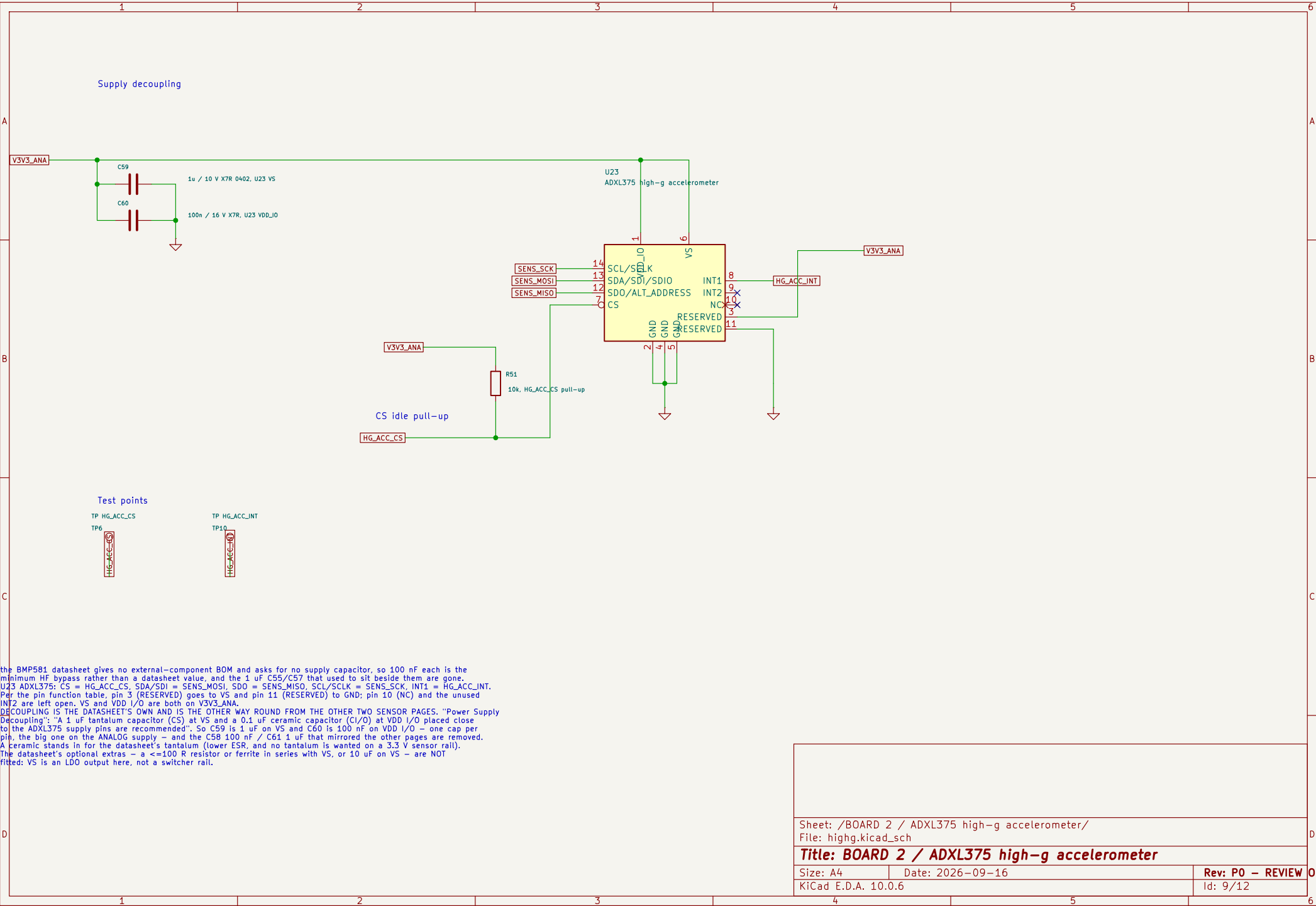
BOARD 2 – U21 ICM-45686 6-AXIS IMU on the shared sensor SPI bus, powered from V3V3_ANA.



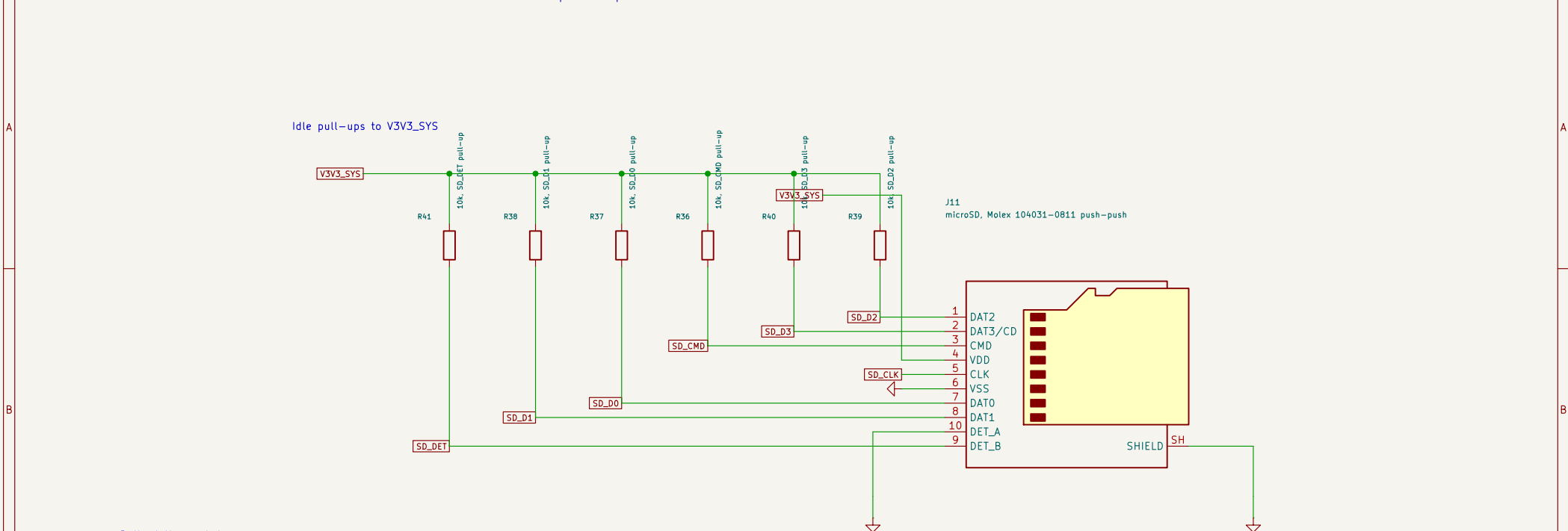
BOARD 2 – U22 BMP581 BAROMETER on the shared sensor SPI bus, powered from V3V3_ANA.



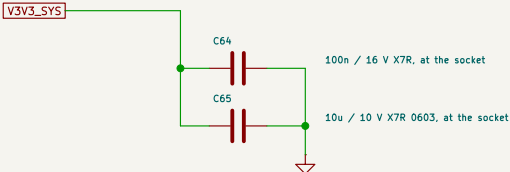
BOARD 2 – U23 ADXL375 HIGH-G ACCELEROMETER on the shared sensor SPI bus, from V3V3_ANA.



VDD = V3V3_SYS¹ with 100 nF + 10 uF at the socket; 10k idle pull-ups on CMD², DAT0-3 and card detect.



Bulk at the socket



J11: DET_A to GND and DET_B to SD_DET with a 10k pull-up, so the input reads low with a card inserted. The shield goes to GND, 100 nF + 10 V (C64/C65) sit at the socket (DESIGN_SPEC "Ralls": SD transients stay on the buck rail); the 47 uF C66 that used to be the third one is removed – a 200 mA write burst on a rail the AP6302 holds with 44 uF nominal of its own COUT does not need an 0805 bulk cap at the socket as well. 10k pull-ups on CMD and DAT0–DAT3 are the SD–standard idle bias and also keep DAT3/CD high for 4-bit mode. VDD = V3V3_SYS.

Sheet: /BOARD 3 / microSD socket/
File: sd.kicad_sch

Title: BOARD 3 / microSD socket

Size: A4

Date: 2026-09-16

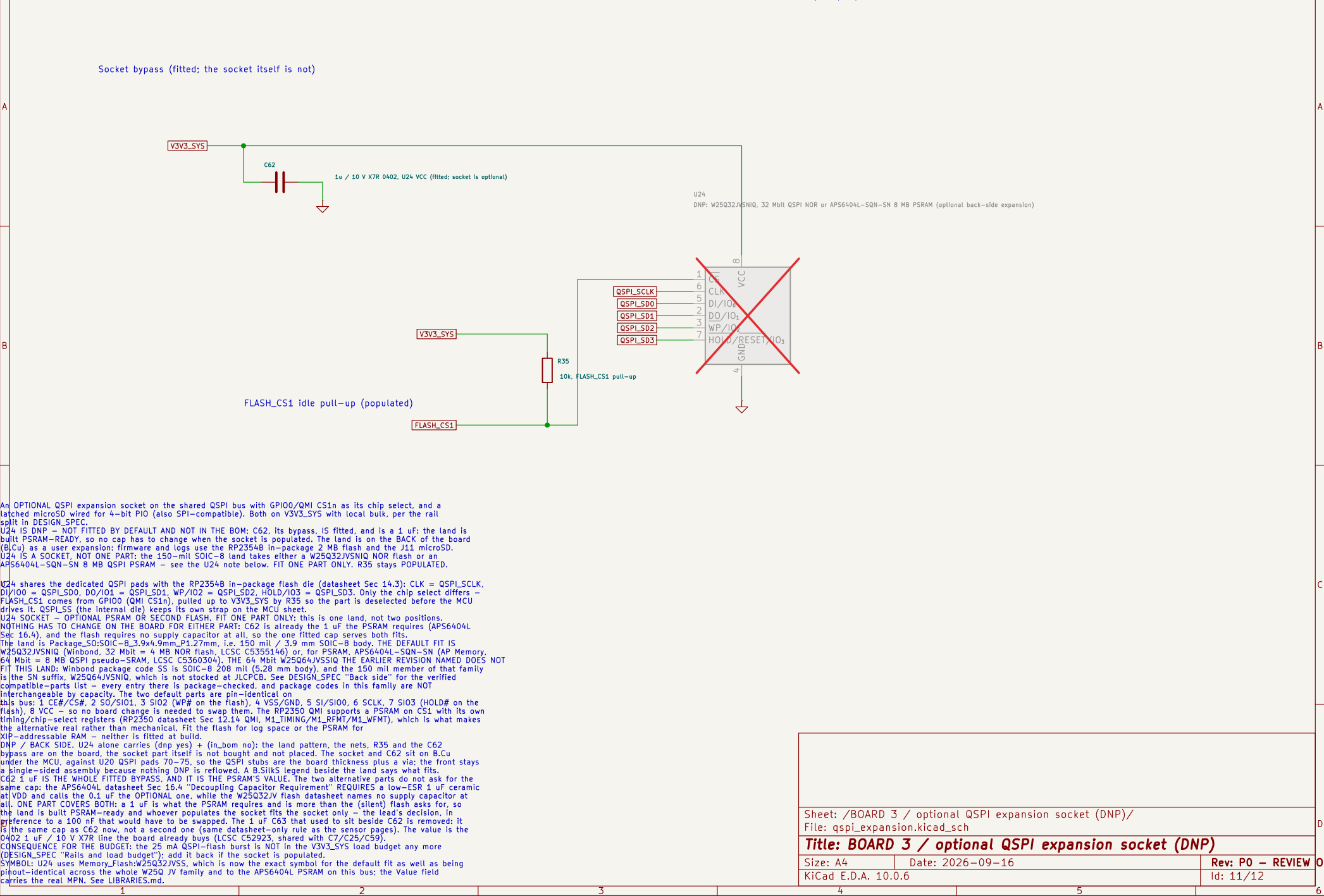
Rev: P0 – REVIEW ONLY

KiCad E.D.A. 10.0.6

Id: 10/12

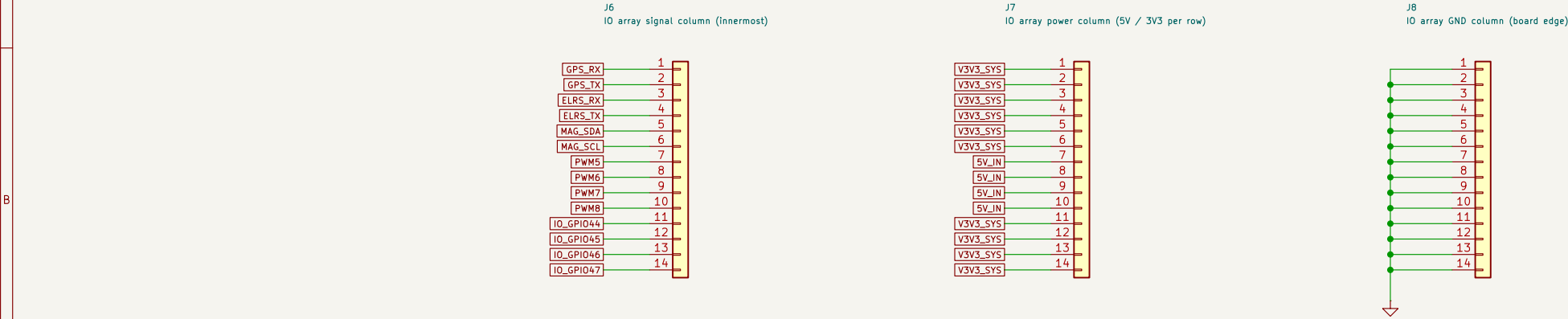
BOARD 3 – U24 OPTIONAL QSPI EXPANSION SOCKET, DNP, ON THE BACK (B.Cu).

One 150-mil SOIC-8 land: fit EITHER a W25Q32JVSNQ NOR flash OR an APS6404L-SQN-SN PSRAM. R35 stays populated.

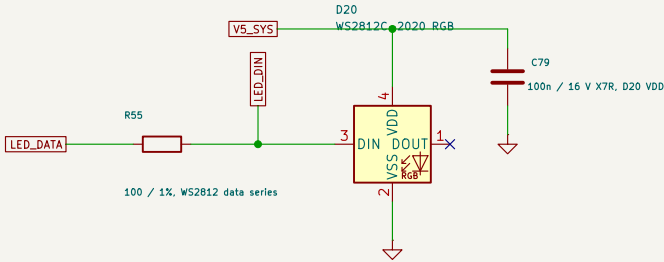


BOARD 4 – THE 3 x 14 IO BLOCK (J6 SIGNAL / J7 POWER / J8 GND), THE WS2812C STATUS LED AND THE FOUR M3 GROMMET HOLES. Every signal that leaves the board except the ESC row J3 and the DBG pads J10 leaves here.

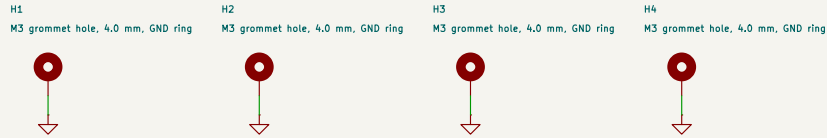
IO array – one row per module lead



WS2812C status LED



Mechanical



IO ARRAY – ONE 3–COLUMN x 14–ROW THROUGH–HOLE BLOCK ON THE RIGHT EDGE, 2.54 mm, madflight FC3v2 style. It replaces the old 2x16 left–edge array AND the J12/J13/J14 servo block: every signal that leaves this board except the ESC row (J3) and the SWD pads (J10) now leaves on this one block. COLUMN RULE, from the board edge inward: GND (J8) | POWER (J7) | SIGNAL (J6). The SIGNAL column is the INNERMOST one, next to the MCU, so the 14 signal traces are the short ones; the two rails run down the outside where a plane stitch is cheap. A 3–wire module lead (signal, power, ground) plugs straight across ONE row. THREE Conn_01x14 SYMBOLS, not one 3x14: no 3–row generic connector symbol ships with this KiCad install, and three 1x14 rows with a body–trimmed courtyard about exactly on the 2.54 mm grid (see tools/setup_pcb.py). DS–009 NAMING IS KEPT: a port label is named for the PERIPHERAL pin it belongs to, so GPS_RX / ELRS_RX are module inputs driven by the RP2354B UART TX pins (rows 1 and 3, silk T0 / T1) and GPS_TX / ELRS_TX are module outputs that land on the MCU UART RX pins (rows 2 and 4, silk R0 / R1).

PWM2 = GPIO39
PWM3 = GPIO38
PWM4 = GPIO36

RAILS ON THE ARRAY: rows 1–6 (GPS, ELRS, magnetometer) and rows 11–14 take V3V3_SYS; rows 7–10 (servos S5–S8) take 5V_IN. Every row has its own GND pin in the J8 column, so a signal return is never more than 2.54 mm away and no row can be mis–plugged into a neighbour’s ground. THE FOUR 5 V SERVO ROWS ARE ON 5V_IN, AHEAD OF THE OR–ING FET (DESIGN_SPEC “Decisions”, 2026–09–17). 5V_IN is the external BEC on J3 pin 7, so servo and module current goes from the BEC straight to the rows and never crosses Q1: the FET carries only the two 3.3 V regulators and the status LED, and a USB host never sees the servo rows at all. The rows are therefore NOT powered from USB – plugging in USB alone leaves every module and servo dead, which is the intended bench behaviour and the reverse of the previous revision. CURRENT: rows 7–10 count against the external BEC; rows 1–6 and 11–14 count in the V3V3_SYS budget. No fuse and no per–row current limit: the array is a system–integration connector, not a protected port. NO LOCAL BULK AT THE ARRAY ANY MORE: C20/C21 (10 uF on 5V_IN) and C22 (10 uF on V3V3_SYS) are removed. The 5 V rows are fed by the external BEC through C19 100 uF at the J3 pad row, and the 3.3 V rows by the AP63203 and its 44 uF nominal of COUT: three 0603 at the connector added nothing either rail did not already have, and a module’s own inrush is its own problem – this block is a connector, not a supply.

MECHANICAL: H1–H4 are the four 4.0 mm grommet holes on the standard 30.5 x 30.5 mm pattern, off–centre in X, each now a PLATED hole with a 6.4 mm GND pad ring stitched by eight 0.6/0.3 mm vias, and a 6.5 mm top–side courtyard for the grommet flange. Pin 1 of every hole is GND, so the frame screws bond to the ground plane on every layer. See DESIGN_SPEC “Mounting and orientation”.

IO ARRAY ROW MAP (row n = J6 pin n signal J7 pin n power J8 pin n GND)				
1	T0	GPS_RX	V3V3_SYS	GND UART1 TX -> GPS RX (GPIO24)
2	R0	GPS_TX	V3V3_SYS	GND UART1 RX <- GPS TX (GPIO25)
3	T1	ELRS_RX	V3V3_SYS	GND UART0 TX -> ELRS RX (GPIO16)
4	R1	ELRS_TX	V3V3_SYS	GND UART0 RX <- ELRS TX (GPIO17)
5	SDA	MAG_SDA	V3V3_SYS	GND I2C1 SDA (GPIO22), 4.7k pull–up on the MCU sheet
6	SCL	MAG_SCL	V3V3_SYS	GND I2C1 SCL (GPIO23), 4.7k pull–up on the MCU sheet
7	S5	PWM5	5V_IN	GND PWM2 A (GPIO20), servo 5
8	S6	PWM6	5V_IN	GND PWM1 B (GPIO19), servo 6
9	S7	PWM7	5V_IN	GND PWM7 B (GPIO15), servo 7
10	S8	PWM8	5V_IN	GND PWM2 B (GPIO5), servo 8
11	A44	IO_GPIO44	V3V3_SYS	GND GPIO44 = ADC4, spare
12	A45	IO_GPIO45	V3V3_SYS	GND GPIO45 = ADC5, spare
13	A46	IO_GPIO46	V3V3_SYS	GND GPIO46 = ADC6, spare
14	A47	IO_GPIO47	V3V3_SYS	GND GPIO47 = ADC7, spare

PWM MAP
PWM5 = GPIO20 (row 7, S5)
PWM6 = GPIO19 (row 8, S6)
PWM7 = GPIO15 (row 9, S7)
PWM8 = GPIO5 (row 10, S8)

ON THE ESC PAD ROW (J3, POWER 1)
PWM1 = GPIO43

Sheet: /BOARD 4 / IO array, status LED, mounting holes/
File: io_block.kicad_sch

Title: BOARD 4 / IO array, status LED, mounting holes

Size: A3	Date: 2026–09–16	Rev: P0 – REVIEW ONLY
KiCad E.D.A. 10.0.6		Id: 12/12